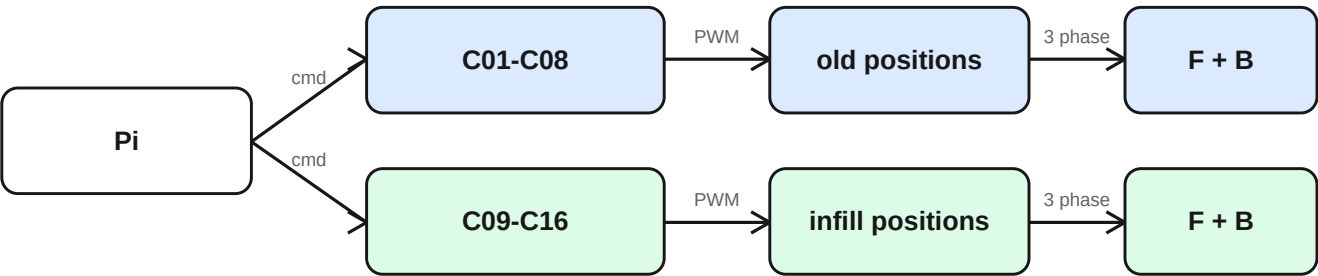


COAXIAL WINDWALL

Wiring Assembly Manual

8x8x2 / 128 motors / two ESC planes

KEEP REAL MOTOR OUTPUT DISABLED UNTIL BENCH TESTS PASS



Operator/front view

P01	P02	P03	P04	P33	P34	P35	P36
P05	P06	P07	P08	P37	P38	P39	P40
P09	P10	P11	P12	P41	P42	P43	P44
P13	P14	P15	P16	P45	P46	P47	P48
P17	P18	P19	P20	P49	P50	P51	P52
P21	P22	P23	P24	P53	P54	P55	P56
P25	P26	P27	P28	P57	P58	P59	P60
P29	P30	P31	P32	P61	P62	P63	P64

Original 8x8 Pico-block numbering

Generated from the harness-preserving project mapping.

C01-C08 reuse old wiring. C09-C16 fill the middle positions. Front/back motors use adjacent channels.

0

Parts, Planes, and Labels

Read the labels before cutting or crimping any harness.

P01-P64

wind pixels

physical 8x8 positions

F01-F64

front motors

Pichler QX-45 ESC plane

B01-B64

back motors

Pichler QX-45 ESC plane

C01-C08

old harness

reuse existing 8x8 cabling

C09-C16

new infill

fill middle positions

CH1-CH8

outputs

Pico GP0-GP7

CTRL-GND

ground

shared signal reference

BEC+

unused

red wire insulated

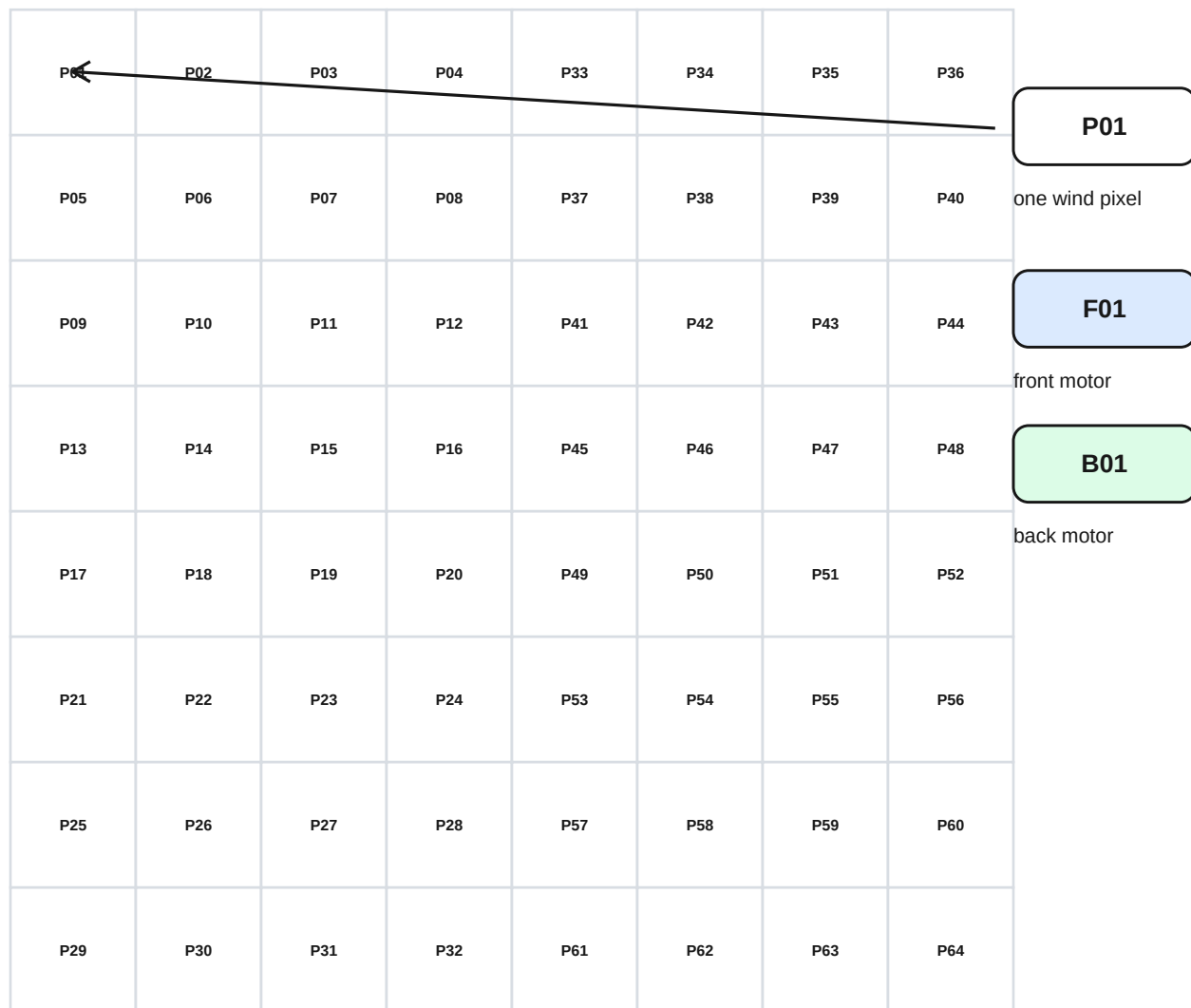
BEC/red wires stay disconnected. Pi/Picos use a separate regulated supply.

1

Label the Frame

Put pixel labels on the frame before unplugging or extending wiring.

Operator/front view



Original 8x8 Pico-block numbering

Front/back are defined from the operator side. Do not swap this later.

2

Mount Controller Banks

C01-C08 reuse old harnesses. C09-C16 fill the middle positions.

Old harness / C01-C08

C01	P01, P04, P05, P08 R1C1, R1C4, R2C1, R2C4
C02	P09, P12, P13, P16 R3C1, R3C4, R4C1, R4C4
C03	P17, P20, P21, P24 R5C1, R5C4, R6C1, R6C4
C04	P25, P28, P29, P32 R7C1, R7C4, R8C1, R8C4
C05	P33, P36, P37, P40 R1C5, R1C8, R2C5, R2C8
C06	P41, P44, P45, P48 R3C5, R3C8, R4C5, R4C8
C07	P49, P52, P53, P56 R5C5, R5C8, R6C5, R6C8
C08	P57, P60, P61, P64 R7C5, R7C8, R8C5, R8C8

New infill / C09-C16

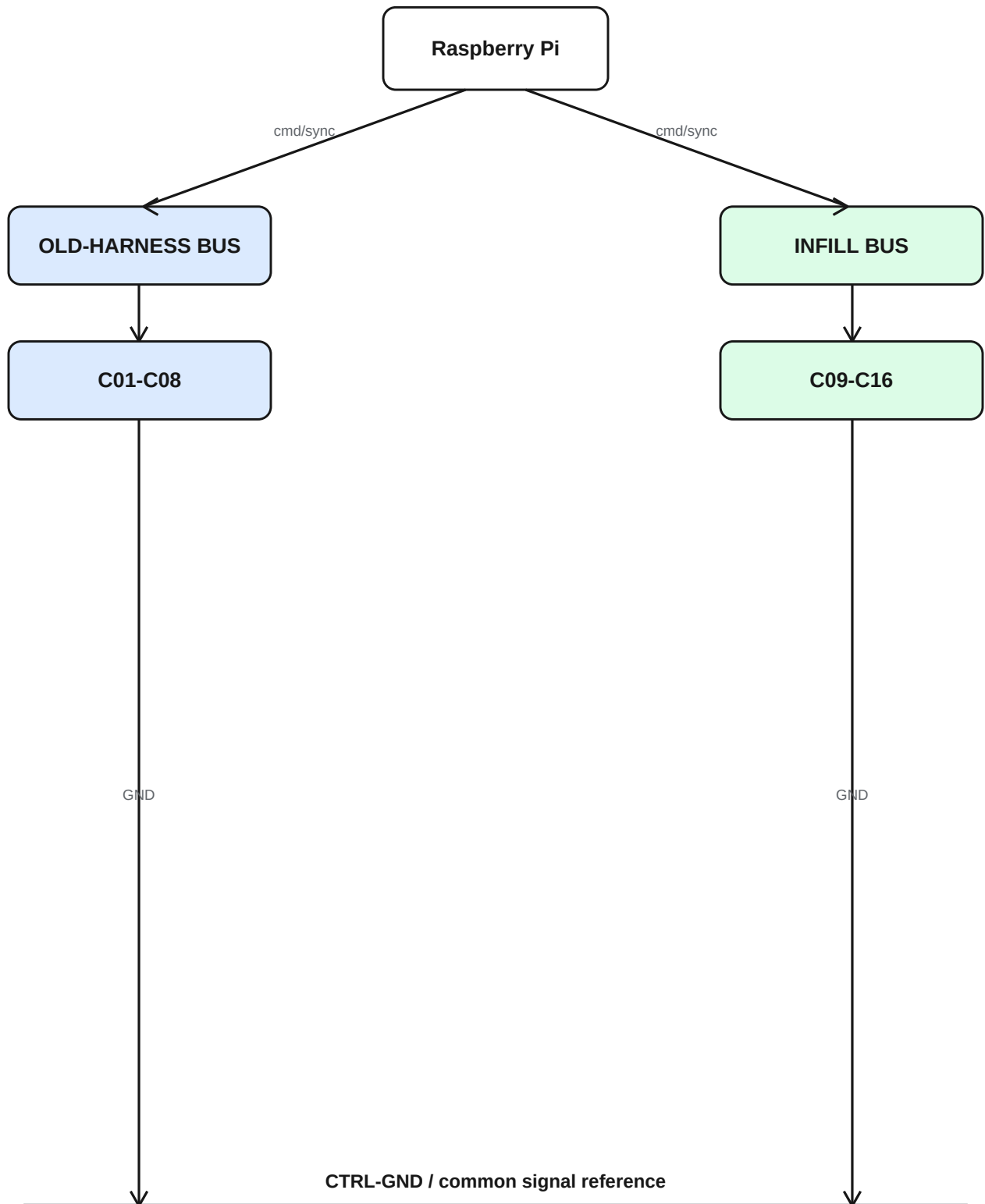
C09	P02, P03, P06, P07 R1C2, R1C3, R2C2, R2C3
C10	P10, P11, P14, P15 R3C2, R3C3, R4C2, R4C3
C11	P18, P19, P22, P23 R5C2, R5C3, R6C2, R6C3
C12	P26, P27, P30, P31 R7C2, R7C3, R8C2, R8C3
C13	P34, P35, P38, P39 R1C6, R1C7, R2C6, R2C7
C14	P42, P43, P46, P47 R3C6, R3C7, R4C6, R4C7
C15	P50, P51, P54, P55 R5C6, R5C7, R6C6, R6C7
C16	P58, P59, P62, P63 R7C6, R7C7, R8C6, R8C7

Mount labels must match docs/MOTOR_CONTROLLER_MAPPING.md before wiring.

3

Wire Control and Ground

The Pi controls both banks. Picos create the ESC PWM outputs.

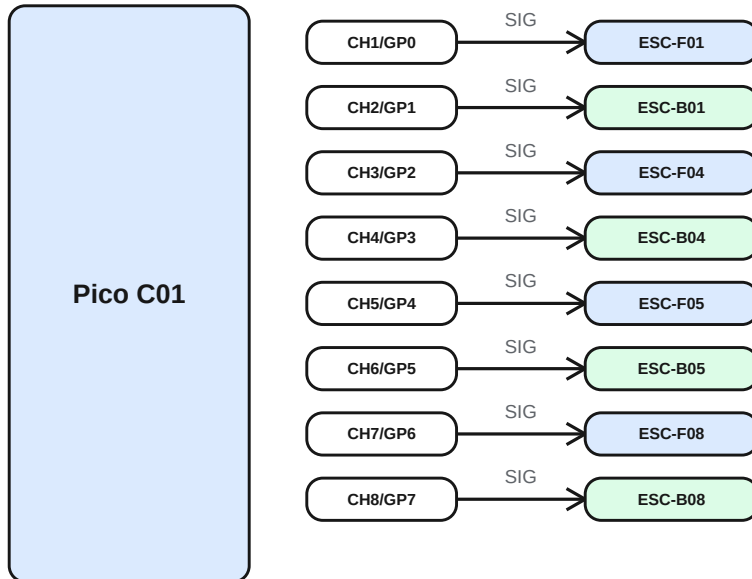


Do not wire 128 ESC signals directly to Pi GPIO. Use the Pico controller outputs.

4

Build the Signal Harness

Every controller repeats the same CH1-CH8 to GP0-GP7 pattern.



Example:

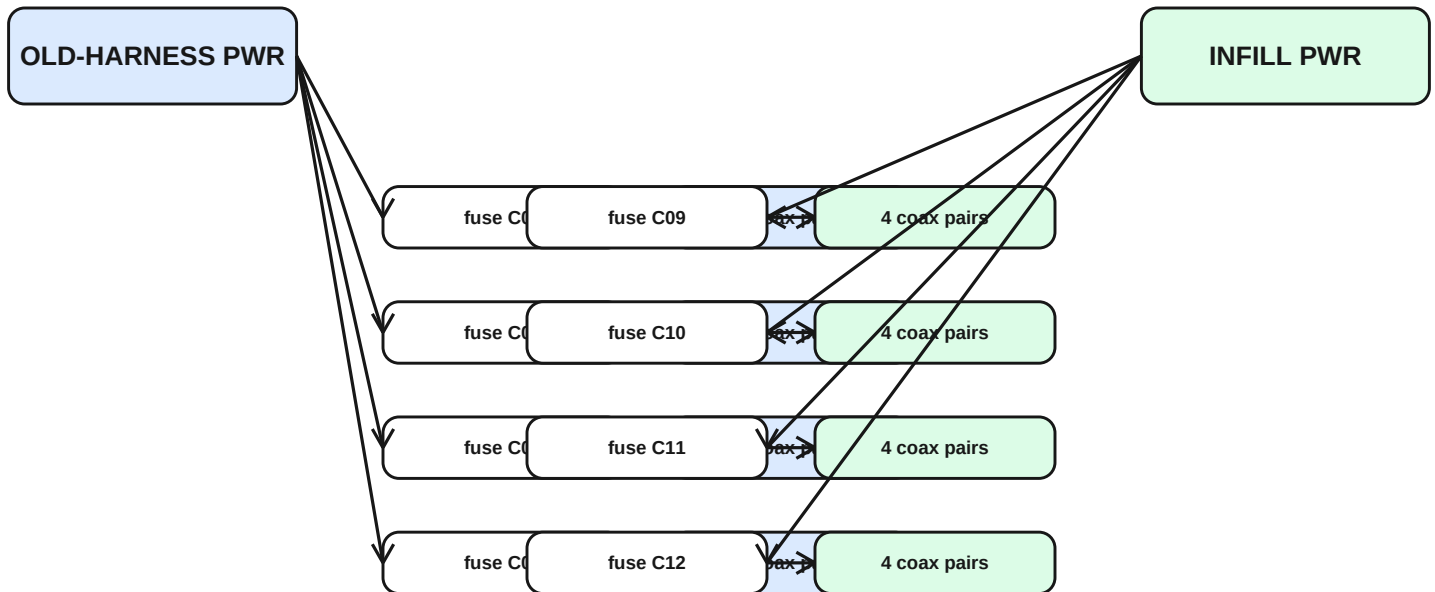
C01 reuses old wiring. Adjacent outputs are front/back for P01, P04, P05, P08.

Each ESC gets its own signal channel. Do not join front and back pair signals.

5

Wire Motor Power

Fuse by practical harness/controller branch, then verify current per branch.



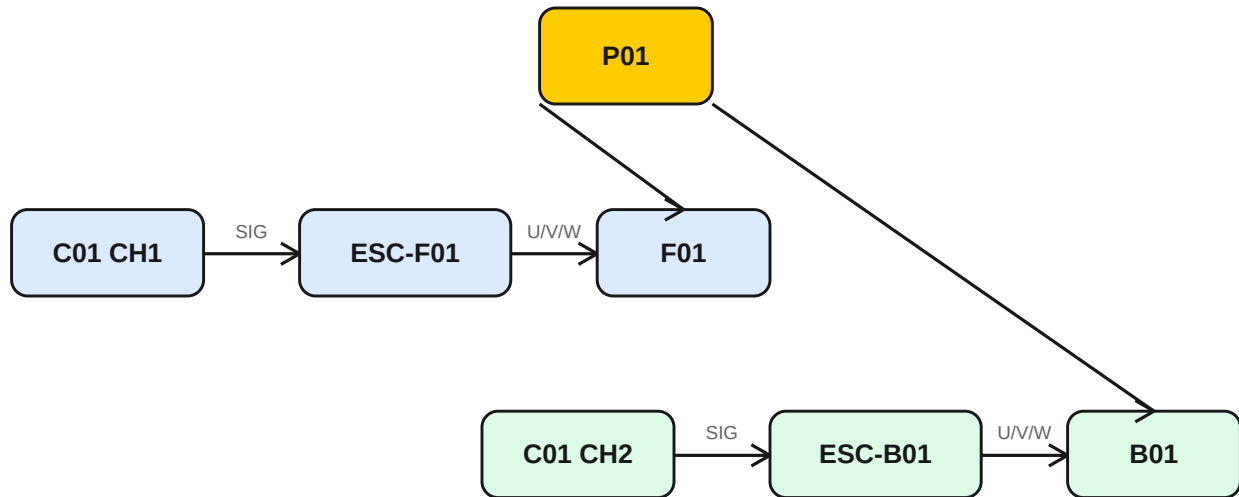
Continue same pattern through C08 and C16.

Fuse from measured current, wire gauge, and connector rating. Do not size only from ESC max amps.

6

Wire One Coaxial Pair

Front and back motors share a pixel label, but use separate ESCs and outputs.



P01 is one air pixel at R1C1. F01 and B01 should be tested as one pair only after each ESC/motor works alone.

Separate signals provide independent front/back control for the QX-45 ESCs.

7

Bring Up Slowly

No full-wall run until each lower-risk stage passes.

- ☐ 1. Label P01-P64, F01-F64, B01-B64, C01-C16.
- ☐ 2. Power Pi/Picos only. Confirm controllers boot.
- ☐ 3. With ESC power off, confirm every channel idles.
- ☐ 4. Bench-test C01 channel order with no live wall load.
- ☐ 5. Bench-test one F motor and one B motor with no live wall load.
- ☐ 6. Calibrate/program all QX-45 ESCs consistently.
- ☐ 7. Test one coaxial pair at low PWM.
- ☐ 8. Test one old-harness controller.
- ☐ 9. Test one new-infill controller.
- ☐ 10. Expand carefully controller by controller.

Emergency stop, idle-on-exit, and idle-on-signal-loss must be proven before real operation.

8

Print These Labels

Use the same words on frame, ESCs, controller outputs, and power branches.

P01

F01 / ESC-F01 / FP / QX-45 / C01-CH1 / GP0

B01 / ESC-B01 / BP / QX-45 / C01-CH2 / GP1

C01-PWR

C09-PWR

CTRL-GND

SIG-GND

BEC+ UNUSED

C01-CH1 SIG -> ESC-F01 SIG

C01-CH2 SIG -> ESC-B01 SIG